User Manual

Introduction

The Drawing Automation Add-In simplifies the process of creating 2D drawings from 3D parts or assemblies in SolidWorks. It allows users to automatically place views, apply dimensions based on preferences, and handle multiple files through batch processing.

Installation Instructions

1. **Prerequisites:**
    - Ensure that SolidWorks is installed on your system.
    - Verify that your SolidWorks version supports .NET add-ins.
2. **Add-In Installation:**
    - Copy the **DrawingAutomation.dll** file to a folder on your system.
    - Open SolidWorks and go to **Tools > Add-Ins**.
    - Click Browse and navigate to the folder where you saved DrawingAutomation.dll.
    - Select the file and click Open.
    - Enable the checkbox next to the add-in in the Add-Ins window to activate it.
3. **Startup Configuration:**
    - To enable the add-in to load automatically on startup, check the Startup box next to the add-in in the Add-Ins window.

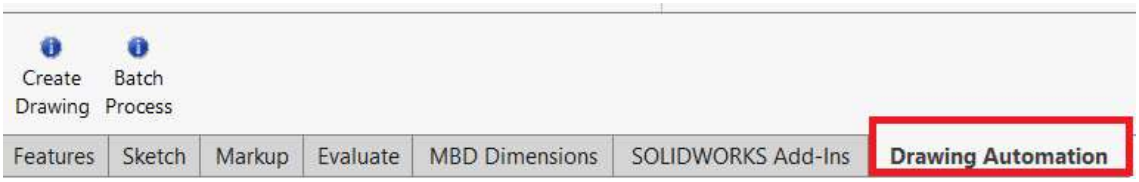
Features

1. **Create Drawing:**
    - Automatically generate a 2D drawing from the active part or assembly.
    - Choose from predefined templates for the drawing sheet size.
    - Place 3rd angle projection views (Front, Top, Isometric).
    - Add dimensions based on user-selected preferences.

2. **Batch Processing:**
    - Process multiple part files to generate drawings in bulk.
    - Select a folder of files, and the add-in will handle the rest.
    - Automatically saves the generated drawings in a specified location.

User Interface

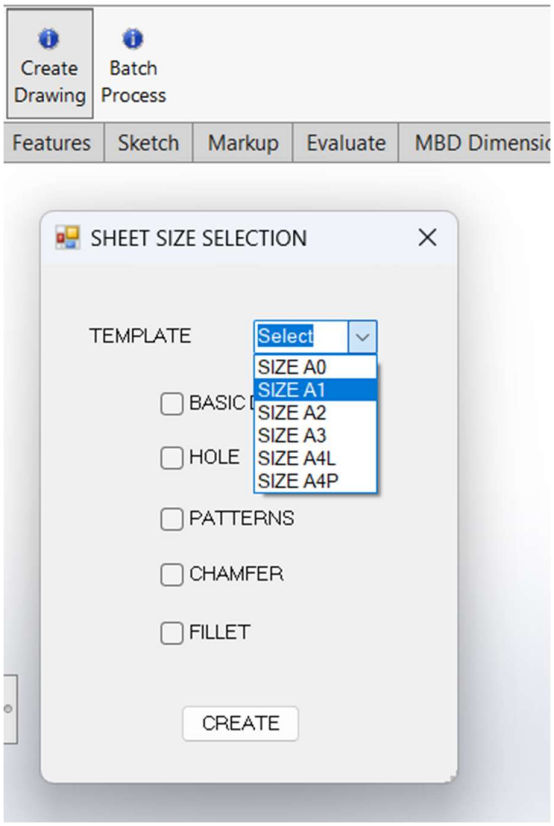
1. **Tab and Buttons:**
    - After installation, a new tab titled **"Drawing Automation"** appears in SolidWorks.

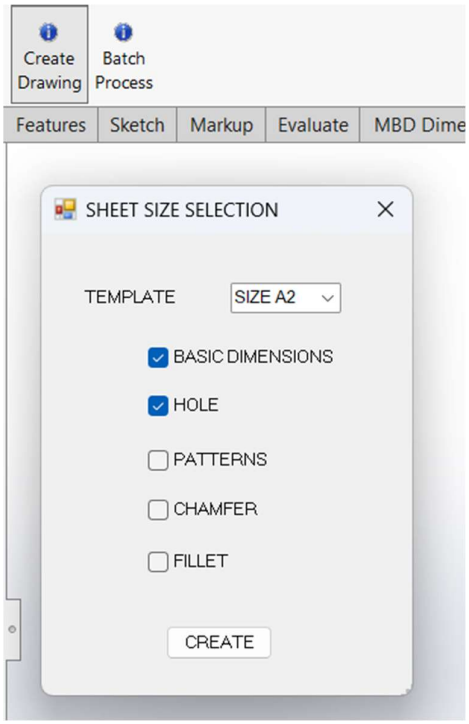


    - Buttons under the tab:
        - **Create Drawing**: Launches the drawing creation process for the active part.
        - **Batch Process**: Opens a dialog to process multiple files for drawing generation.



2. **Template Selection Form:**
    - When creating a drawing, a form prompts the user to select the desired sheet size (e.g., A0, A1, A2).
    - Dimensioning preferences can also be selected, including:
        - Basic Dimensions.
        - Hole Dimensions.
        - Fillet Dimensions.
        - Pattern Dimensions.

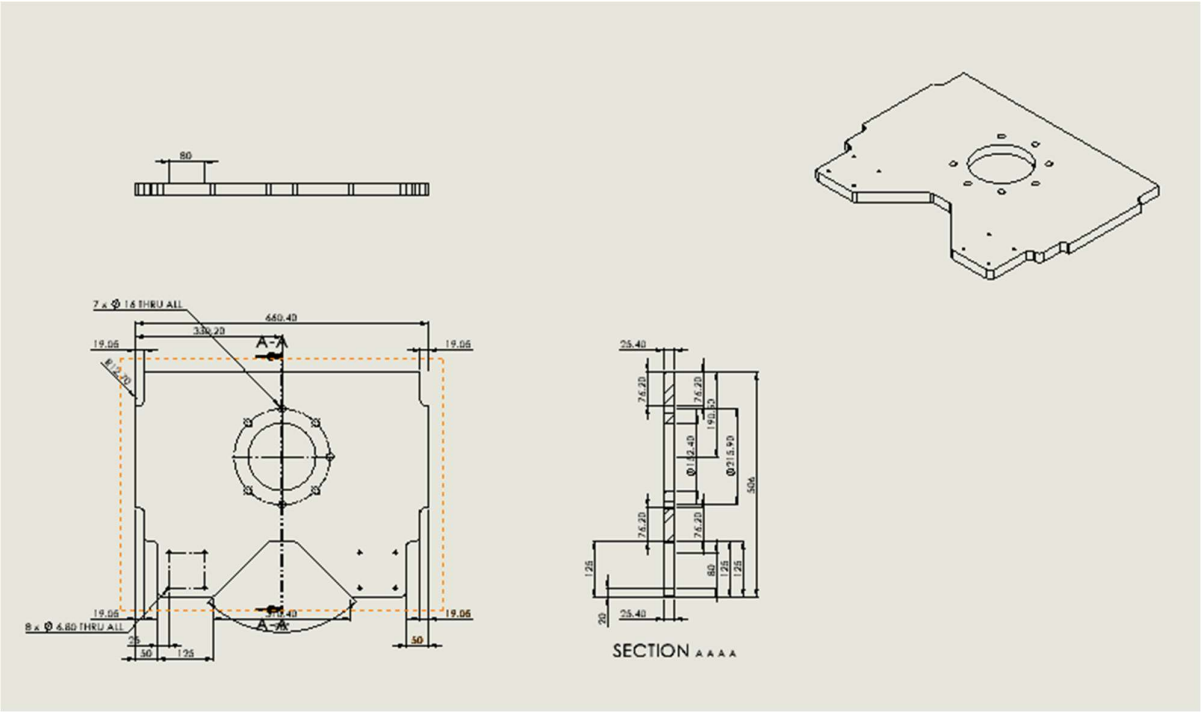


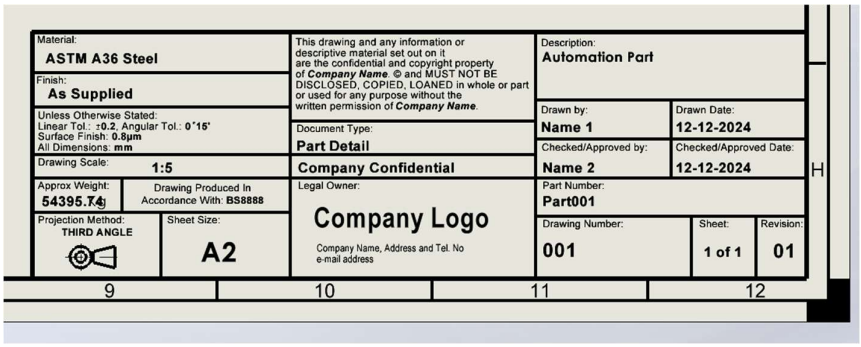


Step-by-Step Guide

1. Creating a Single Drawing

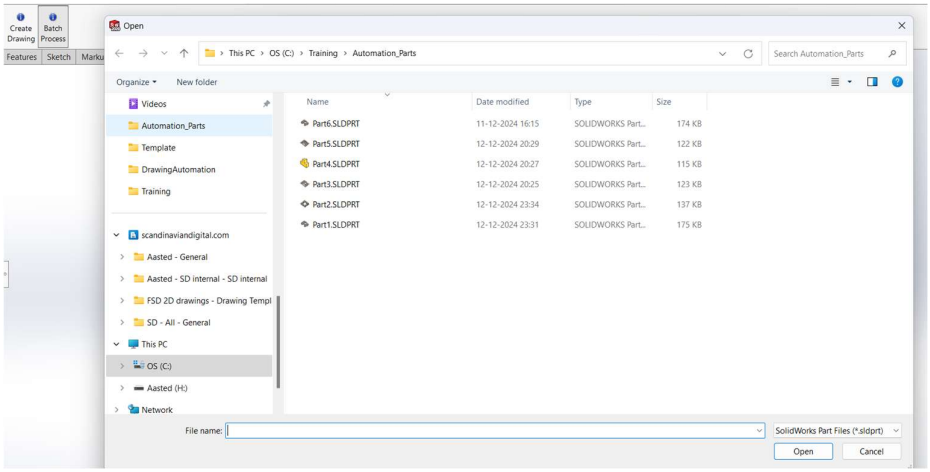
1. Open the desired part or assembly file in SolidWorks.
2. Click on **Create Drawing** in the **Drawing Automation** tab.
3. Select the required sheet size from the form and choose the dimensions you want to include.
4. The add-in generates the drawing with:
    - 3rd angle projection views.
    - Automatically placed dimensions based on your selections.
5. The drawing is displayed in a new document window.
6. Title block with auto-filled metadata

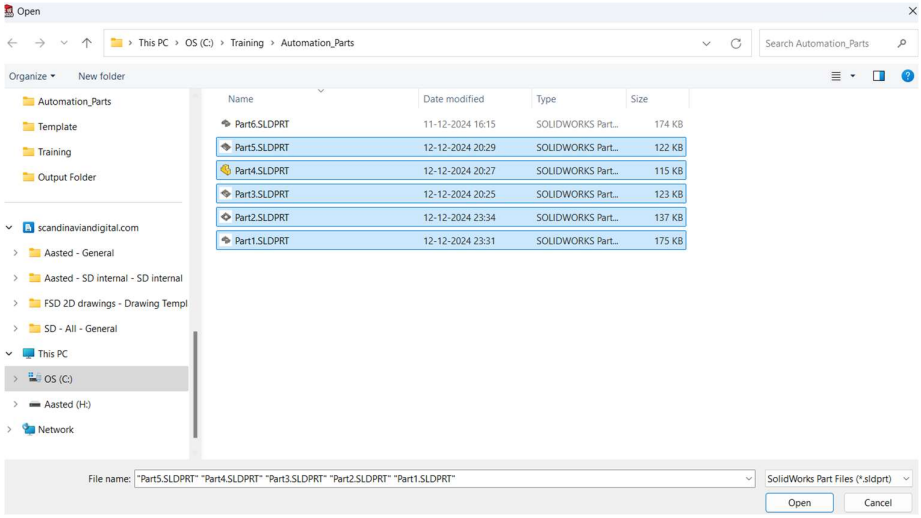


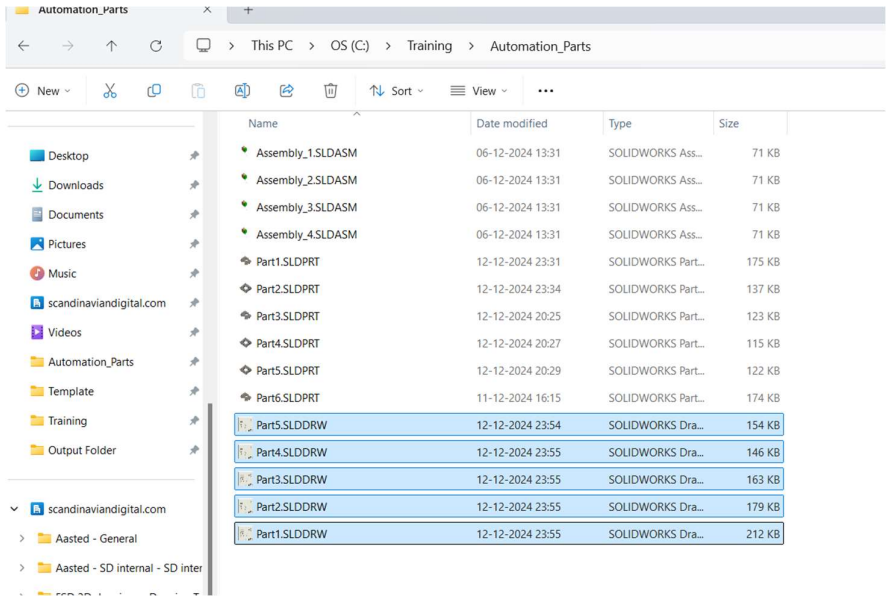


2. Batch Processing for Multiple Files

1. Click on **Batch Process** in the **Drawing Automation** tab.
2. In the file dialog that opens:
    - Select the part files you wish to process.
    - Use Ctrl or Shift for multi-selection.
3. The add-in processes each file:
    - Opens the part file.
    - Creates the drawing using the predefined template.
    - Saves the drawing in the specified output folder.


Create Drawing
Batch Process
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Part5.SLDPRT 12-12-2024 20:29 SOLIDWORKS Part... 122 KB
Part4.SLDPRT 12-12-2024 20:27 SOLIDWORKS Part... 115 KB
Part3.SLDPRT 12-12-2024 20:25 SOLIDWORKS Part... 123 KB
Part2.SLDPRT 12-12-2024 23:34 SOLIDWORKS Part... 137 KB
Part1.SLDPRT 12-12-2024 23:31 SOLIDWORKS Part... 175 KB
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SolidWorks Part Files (*.sldprt)
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Automation_Parts
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Assembly_2.SLDASM 06-12-2024 13:31 SOLIDWORKS Ass... 71 KB
Assembly_3.SLDASM 06-12-2024 13:31 SOLIDWORKS Ass... 71 KB
Assembly_4.SLDASM 06-12-2024 13:31 SOLIDWORKS Ass... 71 KB
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Part1.SLDDRW 12-12-2024 23:55 SOLIDWORKS Dra... 212 KB

Troubleshooting

1. **Add-In Not Visible:**
    - Ensure the add-in is enabled in the Add-Ins manager.
    - Verify that the .dll file is not blocked (Right-click > Properties > Unblock).
2. **Batch Processing Errors:**
    - Ensure all selected files are valid SolidWorks part files (*.sldprt).
    - Verify that the template path and output folder are accessible.
3. **Dimensions Not Placed Correctly:**
    - Some dimensions may not align with the intended view due to feature geometry. Use manual adjustment for final corrections.
4. **File Save Issues:**
    - Check file permissions in the save location.
    - Ensure that SolidWorks has write access to the folder.

Scalability

- The batch process can handle a large number of files, limited only by system resources.
- The modular code design allows easy addition of new features, such as:
    - Custom annotations.
    - Enhanced dimension placement algorithms.